

Types	$\tau ::= \text{bool} \mid \tau_1 \rightarrow \tau_2$
Expressions	$e ::= x \mid \bar{b} \mid \lambda x : \tau. e \mid e_1 e_2$
Values	$v ::= \bar{b} \mid \lambda x : \tau. e$
Contexts	$\Gamma ::= \emptyset \mid (x : \tau) :: \Gamma$
Substitutions	$\sigma ::= \text{id} \mid (v/x) \cdot \sigma$

Figure 1: Syntax

$\frac{\text{Typ-Var}}{\Gamma \ni x : \tau}$	$\frac{\text{Typ-BLit}}{\Gamma \vdash \bar{b} : \text{bool}}$	$\frac{\text{Typ-Abs}}{(x : \tau_1) :: \Gamma \vdash e : \tau_2}$
	$\frac{\text{Typ-App}}{\Gamma \vdash e_1 : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash e_2 : \tau_1}$	
	$\Gamma \vdash e_1 e_2 : \tau_2$	

Figure 2: Syntactic Typing

$\frac{\text{Red-Beta}}{(\lambda x : \tau. e) v \hookrightarrow e[v/x]}$	$\frac{\text{Red-App}_1}{e_1 \hookrightarrow e'_1 \quad e_1 e_2 \hookrightarrow e'_1 e_2}$	$\frac{\text{Red-App}_2}{e_2 \hookrightarrow e'_2 \quad v e_2 \hookrightarrow v e'_2}$
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Figure 3: Reduction

$$\begin{aligned}
\mathcal{V}[\![\text{bool}]\!] &:= \{\bar{b} \mid b \in \mathbb{B}\} \\
\mathcal{V}[\![\tau_1 \rightarrow \tau_2]\!] &:= \{\lambda x : \tau_1. e \mid \forall v. v \in \mathcal{V}[\![\tau_1]\!] \rightarrow e[v/x] \in \mathcal{E}[\![\tau_2]\!]\} \\
\mathcal{E}[\![\tau]\!] &:= \{e \mid \exists v. e \hookrightarrow^* v \wedge v \in \mathcal{V}[\![\tau]\!]\} \\
\mathcal{G}[\![\emptyset]\!] &:= \{\text{id}\} \\
\mathcal{G}[\![x : \tau :: \Gamma]\!] &:= \{(v/x) \cdot \sigma \mid \sigma \in \mathcal{G}[\![\Gamma]\!] \wedge v \in \mathcal{V}[\![\tau]\!]\}
\end{aligned}$$

Figure 4: Logical Relation

$$\Gamma \models e : \tau := \forall \sigma. \sigma \in \mathcal{G}[\![\Gamma]\!] \rightarrow e[\sigma] \in \mathcal{E}[\![\tau]\!]$$

Figure 5: Semantic Typing